

Limits. Part 1

Limits describe how functions behave near specific points. The concept of “Limit” it is invented as a tool to understand trends, holes and instantaneous change. This allow to,

- Define instantaneous rates of change (Derivatives)
- Calculate accumulated change (integrals)
- Analyze function behaviour near discontinuities or asymptotes

The derivative can be think of as a “Instantaneous velocity”, which is the limit of your average velocity as the time interval shrinks to zero. Without limits, we can only talk about average speed over an interval or approximate areas.

For example, let’s assume that a ball is dropped from rest and its distance fallen is $d(t) = 4.9t^2$ meters. Let’s try to compute the speed without limits for the time $t = 3$.

$$s(t) = \frac{d(t + \Delta t) - d(t)}{\Delta t},$$

where Δ is the time interval that we are considering. The average speed is only an estimate. The ratio becomes closer to 29.4 m s^{-1} as the time between them gets shorter⁷, but we can never use this method to find the exact speed at $t=3$. Why? The formula (change in d) / (change in t) needs two points and an interval. At the single moment $t = 3$, there is no change in t , and dividing by zero is not possible. We can only make better and better guesses, but we can’t say what the exact amount is that they are getting closer to.

On the other hand, by using limits, we can get the exact value of the instantaneous velocity, as follows, Starting from the previous average speed equation, however replacing the arguments of the distance function by $3 + h$ and 3, we get,

$$s(t) = \frac{d(3 + h) - d(3)}{h}.$$

h now represents the time interval, and we are going to take the limit as this time interval goes to 0. Next, we evaluate the distance function at the given arguments⁸,

$$s(t) = \frac{29.4h + 4.9h^2}{h}.$$

Since $h \neq 0$, because we are going to take the limit approaching 0,

$$s(t) = 29.4 + 4.9h.$$

Now, we compute the limit as h tends to zero,

$$\begin{aligned} \lim_{h \rightarrow 0} s(t) &= \lim_{h \rightarrow 0} [29.4 + 4.9h] \\ &= 29.4. \end{aligned}$$

Hence, the instantaneous velocity at $t = 3$ is 29.4 m s^{-1} .

January 21

Solutions of previous class activity:

$$f(x) = e^x, \quad g(x) = -(x-1)^2 + 1$$

$$h(x) = e^{-(x-1)^2+1}$$

$$j(x) = -(e^x - 1)^2 + 1$$

$$h^{-1}(x) = \pm \sqrt{-\ln[x] + 1} + 1$$

$$\frac{(g \circ j)(x)}{2} = \frac{-(e^x - 1)^4 + 1}{2}$$

Interval	Avg. Speed	Result
[3, 4]	$\frac{d(4)-d(3)}{1}$	34.3 m/s
⁷ [3, 3.5]	$\frac{d(3.5)-d(3)}{0.5}$	31.85 m/s
[3, 3.1]	$\frac{d(3.1)-d(3)}{0.1}$	29.89 m/s
[3, 3.01]	$\frac{d(3.01)-d(3)}{0.01}$	29.449 m/s
[3, 3.001]	$\frac{d(3.001)-d(3)}{0.001}$	29.4049 m/s

$$d(t) = 4.9t^2 \text{ m}$$

$$d(3) = 44.1 \text{ m}$$

$$d(4) = 78.4 \text{ m}$$

$$d(3.5) = 60.025 \text{ m}$$

$$d(3.1) = 47.089 \text{ m}$$

⁸

$$\begin{aligned} s(t) &= \frac{4.9(3+h)^2 - 4.9(3)^2}{h} \\ &= \frac{4.9(9 + 6h + h^2) - 4.9(9)}{h} \\ &= \frac{44.1 + 29.4h + 4.9h^2 - 44.1}{h} \\ &= \frac{29.4h + 4.9h^2}{h}. \end{aligned}$$